

Tyler Trotter

Exam 4 Foundations of Analysis I Math 3210
I will not use any outside resource on this exam.

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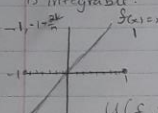
Problem #1:

Using the def. of Integrability (or an equivalent integrability criteria) prove that the function $f(x) = x$ is integrable on $[-1, 1]$ and that $\int_{-1}^1 x dx = 0$.

A function is integrable if there exists a partition P such that

$$\lim_{n \rightarrow \infty} (U(f, P_n) - L(f, P_n)) = 0.$$

Since $f(x) = x$ is monotone on the interval and continuous, it is integrable. Proceed by computing the upper and lower sums.

 Consider a partition which divides the interval $[-1, 1]$ into n equal lengths or sub-intervals. The upper and lower sums are:

$$U(f, P_n) = \sum_{k=0}^{n-1} M_k (x_{k+1} - x_k)$$

$$L(f, P_n) = \sum_{k=0}^{n-1} m_k (x_{k+1} - x_k)$$

We want to show $\lim_{n \rightarrow \infty} (U(f, P_n) - L(f, P_n)) = 0$.

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$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$



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Problem #1, continued.

Split the interval into two equal lengths, both containing 0.

$[-1, 0]$ and $[0, 1]$.

For the second interval, choose a partition such that we divide $[0, 1]$ into $\frac{n}{2}$ equal lengths.

$$U(S, P_n) = \sum_{k=1}^{\frac{n}{2}} \left(S(k) \cdot \left(\frac{1}{n} \right) \right) = \sum_{k=1}^{\frac{n}{2}} \frac{k}{n} \cdot \frac{1}{n} = \frac{1}{n^2} \sum_{k=1}^{\frac{n}{2}} k$$

Using the given formula, we get $\frac{1}{n^2} \sum_{k=1}^{\frac{n}{2}} k = \frac{n(n+1)}{2n^2}$

$$L(S, P_n) = \sum_{k=1}^{\frac{n}{2}} \left(S(k-1) \cdot \left(\frac{1}{n} \right) \right) = \sum_{k=1}^{\frac{n}{2}} \frac{k-1}{n} \cdot \frac{1}{n} = \frac{1}{n^2} \sum_{k=1}^{\frac{n}{2}} (k-1) = \frac{n(n-1)}{2n^2}$$

$$\lim (U(S, P_n) - L(S, P_n)) = \lim \left(\frac{n(n+1) - n(n-1)}{2n^2} \right)$$

$$= \lim \frac{2n}{2n^2} = \lim \frac{1}{n} = 0.$$

For the other interval, $[-1, 0]$, we also divide into $\frac{n}{2}$ equal lengths.

$$U(S, P_n) = \sum_{k=1}^{\frac{n}{2}} S(k) \left(\frac{1}{n} \right) = \sum_{k=1}^{\frac{n}{2}} \left(-\frac{k}{n} \right) \left(\frac{1}{n} \right) = -\frac{1}{n^2} \sum_{k=1}^{\frac{n}{2}} k = -\frac{n(n+1)}{2n^2}$$

$$L(S, P_n) = \sum_{k=1}^{\frac{n}{2}} S(k-1) \left(\frac{1}{n} \right) = \sum_{k=1}^{\frac{n}{2}} \left(-\frac{k-1}{n} \right) \left(\frac{1}{n} \right) = -\frac{1}{n^2} \sum_{k=1}^{\frac{n}{2}} (k-1) = -\frac{n(n-1)}{2n^2}$$

$$\lim (U(S, P_n) - L(S, P_n)) = \lim \left(\frac{-n(n+1) - (-n(n-1))}{2n^2} \right) = \lim \frac{2n}{2n^2} = 0.$$

So, $S(x)$ is integrable on the entire interval. Now we can use any sum and add the intervals up. We get

$$\int_0^1 x dx = \lim_{n \rightarrow \infty} U(S, P_n) = \lim_{n \rightarrow \infty} \left(\frac{n(n+1)}{2n^2} \right) = \frac{1}{2}$$

$$+ \int_{-1}^0 x dx = \lim_{n \rightarrow \infty} U(S, P_n) = \lim_{n \rightarrow \infty} \left(-\frac{n(n+1)}{2n^2} \right) = -\frac{1}{2} \quad \left| \frac{1}{2} + \left(-\frac{1}{2} \right) = 0 \right|$$

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Problem #2:

Prove that $\lim_{n \rightarrow \infty} \int_{1/2}^1 \cos^n(x) dx = 0$

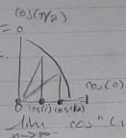
we proved in the homework that we could bring limits in/out of integrals without impacting the result.

Sol. 1. $\int_{1/2}^1 \lim_{n \rightarrow \infty} \cos^n(x) dx$

Consider the unit circle.

$0 < \cos(1) < \cos(1/2) < 1$

Then we consider $\lim_{n \rightarrow \infty} \cos^n(1)$ and $\lim_{n \rightarrow \infty} \cos^n(1/2)$



Since $0 < \cos(1) < 1$ and $0 < \cos(1/2) < 1$, we have a positive fraction, for each of $\cos(1)$ and $\cos(1/2)$. Due to this, I am able to assert that $\cos^n(1) = [\cos(1)]^n$ and $\lim_{n \rightarrow \infty} [\cos(1)]^n \rightarrow 0$ because $\cos(1)$ or $\cos(1/2)$ are represented as follows $(\frac{a}{b})^n$ where $a < b < 1$. This means $a < b$. Then $\lim_{n \rightarrow \infty} (\frac{a}{b})^n = \lim_{n \rightarrow \infty} \frac{a^n}{b^n}$ which goes to 0 because $a < b$ ($a^n < b^n$ means b^n grows much faster).

Now, for any $x \in (1/2, 1)$, $\lim_{n \rightarrow \infty} \cos^n(x) \rightarrow 0$. So, we have

$\int_{1/2}^1 0 dx = 0 \Big|_{1/2}^1 = 0$

Problem #3:

Consider the power series: $\sum_{k=1}^{\infty} \frac{(x-5)^k}{3^k k}$
 a) Find the interval of convergence for this series.

Use Ratio Test: $r = \lim \left| \frac{(x-5)^k (x-5)}{3 \cdot 3^k (k+1)} \cdot \frac{3^k k}{(x-5)^k} \right|$

Continue here:

$x \in (2, 8)$: Check endpoints:

Check

$x=2$:

$\sum \frac{(2-5)^k}{3^k k} = \sum \frac{(-3)^k}{3^k k} = \sum \frac{(-1)^k}{k}$

which converges

conditionally (alternating series test)

Check

$x=8$:

$\sum \frac{(8-5)^k}{3^k k} = \sum \frac{3^k}{3^k k} = \sum \frac{1}{k}$

which diverges (harmonic).

So, the interval of convergence is: $x \in [2, 8)$

b) If we define $f(x)$ to be the sum of the series $\sum \frac{(x-5)^k}{3^k k}$ for $x \in I$, what is $f^{(21)}(5)$?

Using Taylor's Theorem: $f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(5)}{k!} (x-5)^k$

Then $f(x) = \sum_{k=0}^{\infty} \frac{(x-5)^k}{3^k k} = \sum_{k=0}^{\infty} \frac{f^{(k)}(5)}{k!} (x-5)^k$

Then $\frac{f^{(21)}(5)}{21!} = \frac{1}{3^{21}}$; Thus $f^{(21)}(5) = \frac{21!}{3^{21}} = \frac{(21-1)!}{3^{21}} = \frac{20!}{3^{21}}$

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Problem #4:

Suppose f has derivatives of all orders on $(-1,1)$ and there exists a constant k such that $|f^{(n)}(x)| \leq n!k$ for all $x \in (-1,1)$, $\forall n \in \mathbb{N}$.

Prove that the Taylor series for f centered at 0 converges to f on $(-1,1)$. Want to show:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x)^n; \text{ given: } |f^{(n)}(x)| \leq n!k$$

$$\text{so we have } \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x)^n \leq \sum_{n=0}^{\infty} \frac{n!k}{n!} (x)^n$$

True because $|f^{(n)}(x)| \leq nk$ for any $x \in (-1,1)$, which contains 0 and thus holds for $|f^{(n)}(0)| \leq nk$.

$$\sum_{n=0}^{\infty} \frac{n!k}{n!} (x)^n = \sum_{n=0}^{\infty} kx^n. \text{ This is a geometric series where } a=k \text{ and } r=x. \text{ Since } x \in (-1,1), \text{ then } |r| < 1, \text{ which implies the geometric series converges.}$$

By comparison test, we have shown that the Taylor series centered at 0 converges to a function f on the interval $(-1,1)$.